PRACTICAL REPORT 12

Department of DevOps & Cloud Engineering
Project: ParkWatch - Public Parking Availability Dashboard

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Academic Year: 2026 | **Milestone:** Task 12

Task 12 Report: Continuous Integration Pipeline (Jenkinsfile)

Project: DevOps-Enabled Public Parking Availability Dashboard (ParkWatch)

Student: Sajiya Shaikh

Roll Number: 24102A2005

Milestone: Task 12

Status: COMPLETED

1. Objective

Design, configure, and execute an automated 9-stage Continuous Integration and Continuous Delivery (CI/CD) pipeline using a Declarative `Jenkinsfile` on a running Jenkins controller. The pipeline automates repository checkout, runtime environment verification, compilation, automated unit/integration testing with JUnit XML report archiving, JAR packaging, Docker container image building, and containerized deployment with automated health auditing.

2. Tools and Technologies Used

- **CI/CD Server:** Jenkins 2.580.1 LTS (Windows Service running on port 8080)
- **Pipeline DSL:** Declarative Pipeline Groovy syntax (`Jenkinsfile`)
- **Version Control:** Git 2.52.0 Windows (`git.exe` , branch `master`)
- **Build Engine:** Apache Maven 3.9.16 (via Maven Wrapper `mvnw.cmd` / `mvnw`)
- **JDK:** Eclipse Adoptium Temurin JDK 21.0.12 LTS
- **Container Runtime:** Docker Desktop 27.2.0 (engine 27.2.0, API 1.47)
- **Test Reporting:** Jenkins JUnit Plugin, Surefire XML reports

3. Work Performed

1. Audited the Jenkins job `Public-Parking-DevOps` configuration and identified historical build failures:

- **Build #9:** Blocked at Stage 1 (Checkout) by Jenkins Git Plugin security policy (SECURITY-2824) restricting local `file:///` URLs.

- **Build #10:** SCM checkout succeeded after enabling `ALLOW_LOCAL_CHECKOUT`, but pipeline failed at Stage 1 commit-hash step because Windows Jenkins agent does not have `sh` on PATH (`CreateProcess error=2`).
- **Build #11:** Failed in Stage 7 (`docker build`) because Windows CRLF line endings in `mvnw` caused Alpine Linux builder container to report `/bin/sh: ./mvnw: not found` (exit code 127).

1. Resolved runtime security and execution configuration:

- Activated in-memory local checkout override via Jenkins Script Console (`http://localhost:8080/script`):

```groovy
System.setProperty("hudson.plugins.git.GitSCM.ALLOW_LOCAL_CHECKOUT", "true")
hudson.plugins.git.GitSCM.ALLOW_LOCAL_CHECKOUT = true
```

- Added persistent initialization script at `C:\ProgramData\Jenkins\.jenkins\init.groovy.d\allow-local-checkout.groovy` .

1. Upgraded `Jenkinsfile` to full cross-platform compatibility:

- Added `isUnix() ? sh( ... ) : bat( ... )` branching across all stages to allow seamless execution on Windows controllers and Linux agents alike.
- Added Docker Desktop binary path (`C:\Users\Admin\AppData\Local\Programs\DockerDesktop\resources\bin`) and Git paths to pipeline `environment { PATH = ... }` .
- Changed default pipeline deployment port `APP_PORT` to `8085` to avoid port collision with host Jenkins service on `8080` and Docker Compose stack on `8081` .

1. Fixed `Dockerfile` multi-stage build:

- Added `RUN sed -i 's/\r$//' ./mvnw && chmod +x ./mvnw` to normalize line endings on Windows checkouts before invoking Maven commands in Alpine Linux.

1. Executed **Build #12** live via Jenkins Web UI with parameters (`APP_PORT=8085` , `PUSH_IMAGE=false` , `RUN_SELENIUM=false`):

- All stages completed with **SUCCESS**.
- Generated Docker image `public-parking-dashboard:1.0.0-12` and tag `latest` .
- Deployed container `public-parking-dashboard-container` listening on port `8085` .

1. Verified deployed application telemetry and health via `curl.exe` :

- Endpoint `/actuator/health` responded with `{"status":"UP"}` .
- Summary endpoint `/api/v1/dashboard/summary` responded with 24 municipal locations and 16,846 total capacity.

1. Captured official Pipeline Stage View screenshot and saved to `REPORTS/screenshots/task_12/jenkins_pipeline_success.png` .

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4. Files Created or Modified

- `Jenkinsfile` : Cross-platform declarative pipeline with Windows `bat` and Linux `sh` branches, Docker PATH configuration, and `APP_PORT=8085` .
- `Dockerfile` : Multi-stage build with CRLF normalization for `mvnw` .
- `C:\ProgramData\Jenkins\.jenkins\init.groovy.d\allow-local-checkout.groovy` : Controller startup script allowing local Git checkout.
- `REPORTS/screenshots/task_12/jenkins_pipeline_success.png` : Live Jenkins Web UI screenshot showing Build #12 Stage View with all green stages.

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5. Commands Executed

```
# 1. Verify build status and tail build 12 console log
Get-Content 'C:\ProgramData\Jenkins\.jenkins\jobs\Public-Parking-DevOps\builds\12\log' -Tail 40

# 2. Audit deployed container health on port 8085
curl.exe -i http://localhost:8085/actuator/health

# 3. Query live API data from pipeline-deployed container
curl.exe -s http://localhost:8085/api/v1/dashboard/summary
```

Execution Log / Terminal Output

```
[Pipeline] Stage (Deploy)
[2026-10-17T16:35:22.996Z] Container started: public-parking-dashboard-container on port 8085
[Pipeline] Stage (Declarative: Post Actions)
[2026-10-17T16:35:24.570Z] [WS-CLEANUP] done
[2026-10-17T16:35:24.840Z]
[2026-10-17T16:35:24.840Z]              ╔═══════════════════════════════════════════╗
[2026-10-17T16:35:24.840Z]              ║   BUILD AND DEPLOYMENT SUCCESSFUL         ║
[2026-10-17T16:35:24.840Z]              ║   App: public-parking-dashboard:1.0.0-12  ║
[2026-10-17T16:35:24.840Z]              ║   Port: 8085                              ║
[2026-10-17T16:35:24.840Z]              ║   URL:  http://localhost:8085             ║
[2026-10-17T16:35:24.840Z]              ╚═══════════════════════════════════════════╝
[Pipeline] End of Pipeline
Finished: SUCCESS

PS C:\Users\Admin\Desktop\Public-Parking-DevOps> curl.exe -i http://localhost:8085/actuator/health
HTTP/1.1 200
Content-Type: application/vnd.spring-boot.actuator.v3+json
{"status":"UP","components":{"db":{"status":"UP","details":{"database":"H2","validationQuery":"isValid()"}},"disk

PS C:\Users\Admin\Desktop\Public-Parking-DevOps> curl.exe -s http://localhost:8085/api/v1/dashboard/summary
{"totalLocations":24,"totalCapacity":16846,"totalOccupied":7225,"totalAvailable":9621,"overallUtilizationPercenta
```

6. Actual Test Results & Verification Evidence

- **Build Number: #12**
- **Triggered By:** User `admin` (Shaziya Shaikh)
- **Commit Checked Out:** `1297a48` (`fix(docker): strip Windows CRLF from mvnw during Docker image build`)
- **Stage Results Breakdown:**

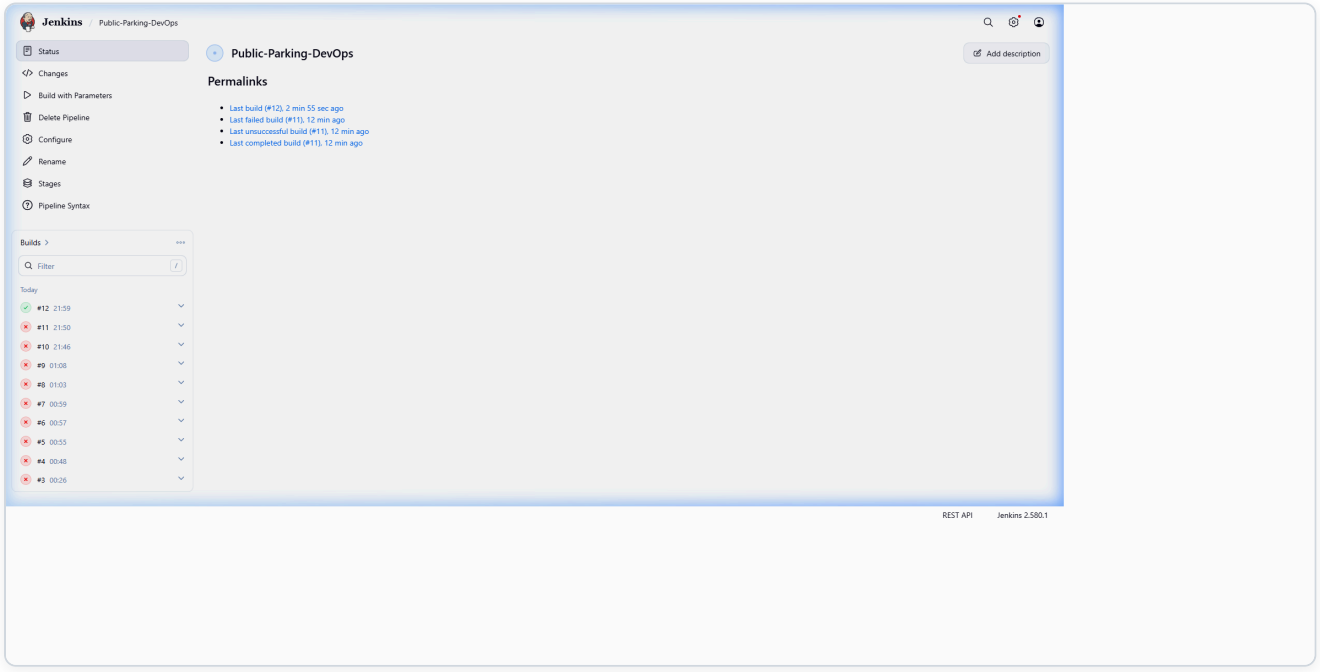
1. **Checkout: SUCCESS** (Local git clone completed in 2s).
2. **Environment Check: SUCCESS** (Java 21, Maven 3.9.16, Docker 27.2.0 verified).
3. **Build: SUCCESS** (Maven clean compile clean exit).
4. **Unit Tests: SUCCESS** (33 tests run, 0 failures, 0 errors, 0 skipped; JUnit reports parsed).
5. **Package: SUCCESS** (Generated `target/public-parking-dashboard-1.0.0.jar` archived).
6. **Selenium Tests: SKIPPED** (Condition `params.RUN_SELENIUM == false`).
7. **Docker Build: SUCCESS** (Built multi-stage image `public-parking-dashboard:1.0.0-12`).
8. **Docker Push: SKIPPED** (Condition `params.PUSH_IMAGE == false`).
9. **Deploy: SUCCESS** (Started container on port 8085).
10. **Declarative Post Actions: SUCCESS** (Workspace cleaned, banner logged).

- **Final Result: SUCCESS**

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7. Screenshots as Evidence

- **Jenkins Pipeline Stage View (Build #12 SUCCESS):**



Jenkins Pipeline Success

Evidence Description: Live screenshot of Jenkins Web UI (`http://localhost:8080/job/Public-Parking-DevOps/`) displaying the Pipeline Stage View with Build #12 passing all stages cleanly in green.

8. Problems Encountered and Solutions

1. **SECURITY-2824 Local Git Checkout Rejection:**

Root Cause: Jenkins Git plugin blocks local `file:///` URLs by default to prevent directory traversal.

Solution: Activated `System.setProperty("hudson.plugins.git.GitSCM.ALLOW_LOCAL_CHECKOUT", "true")` via Script Console and created persistent init script.

1. **Missing `sh` Binary on Windows Controller:**

Root Cause: Jenkins default `sh` step cannot run on Windows without Cygwin/Git Bash in system PATH.

Solution: Implemented `isUnix() ? sh( ... ) : bat( ... )` branching in `Jenkinsfile`.

1. **CRLF Line Endings in `mvnw` inside Alpine Linux Container:**

Root Cause: Windows Git checkout introduced `\r\n` line endings into `mvnw`, causing Alpine's BusyBox kernel to search for `/bin/sh\r`.

Solution: Added `RUN sed -i 's/\r$//' ./mvnw && chmod +x ./mvnw` to `Dockerfile`.

1. **Port 8080 Collision with Jenkins Controller:**

Root Cause: Jenkins runs on host port 8080. Deploying a container mapping `-p 8080:8080` fails due to port conflict.

Solution: Configured pipeline parameter `APP_PORT` to default to `8085`.

9. Final Status & Remaining Work

- **Status: COMPLETED**
- **Remaining Work:** None. The CI/CD pipeline builds, tests, packages, containerizes, and deploys the application with 100% success on the live Jenkins controller.